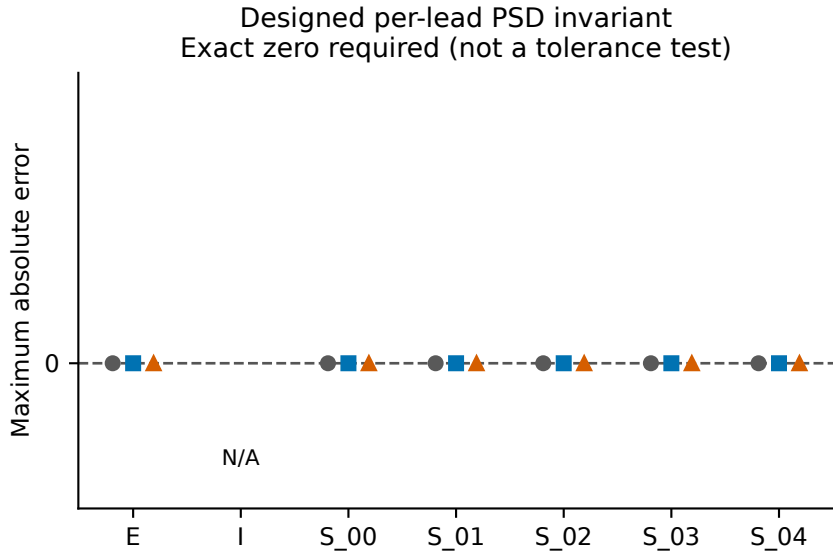
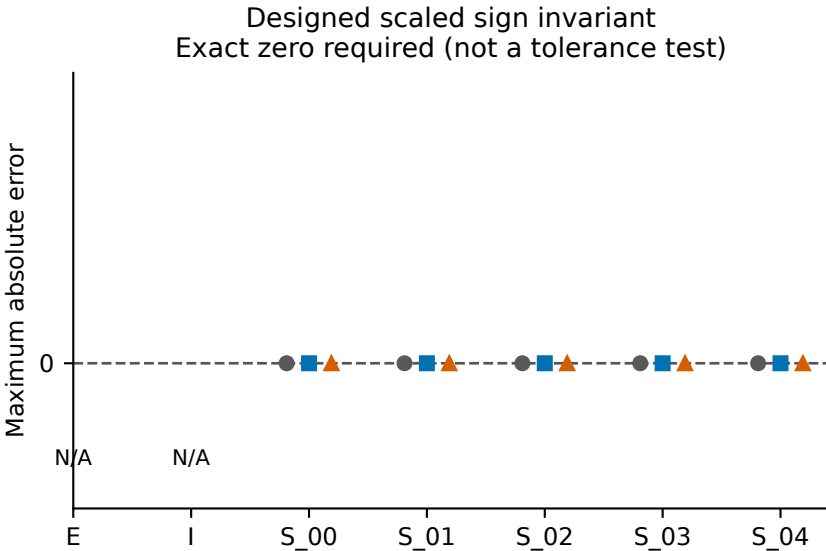
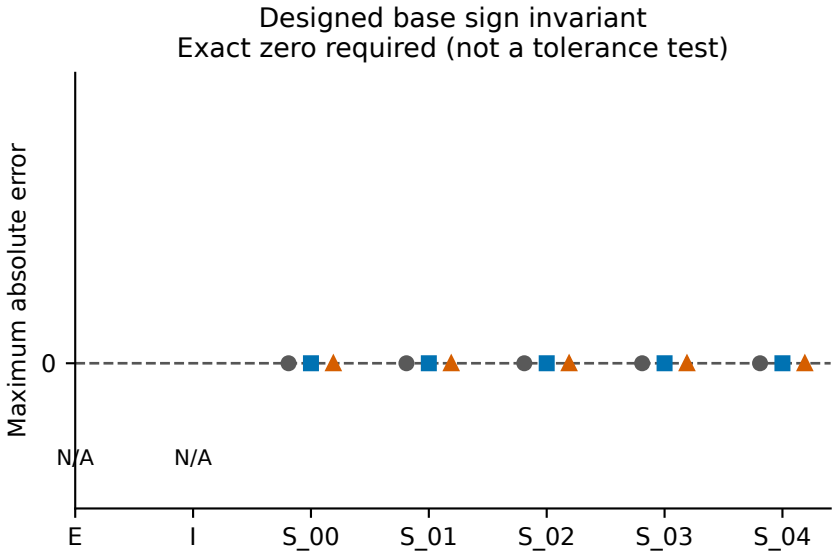
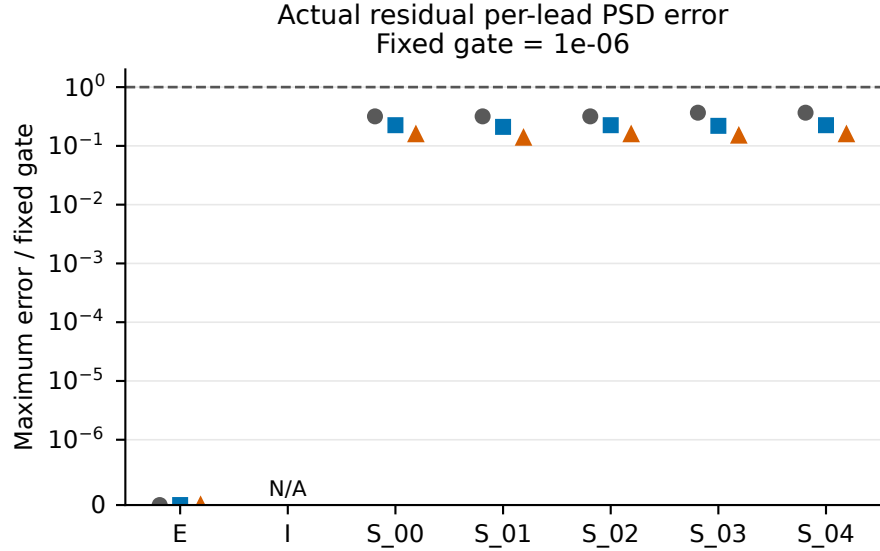
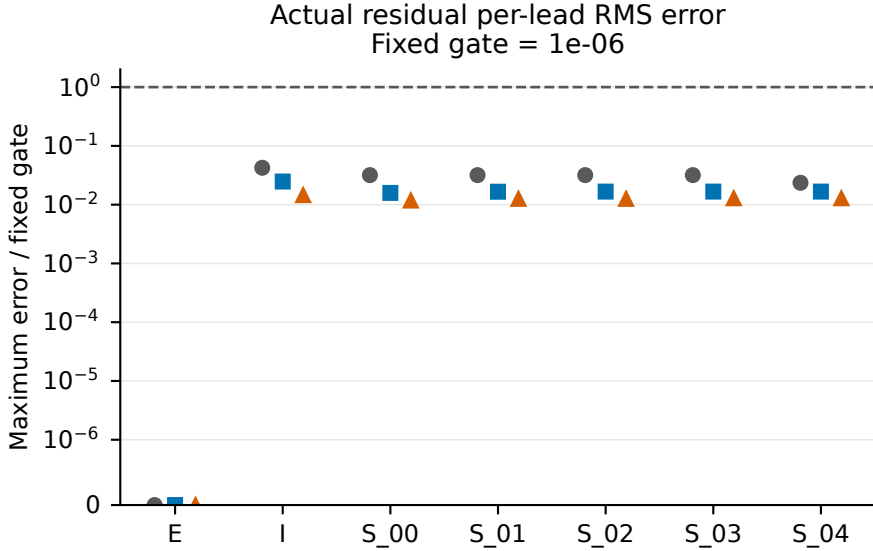
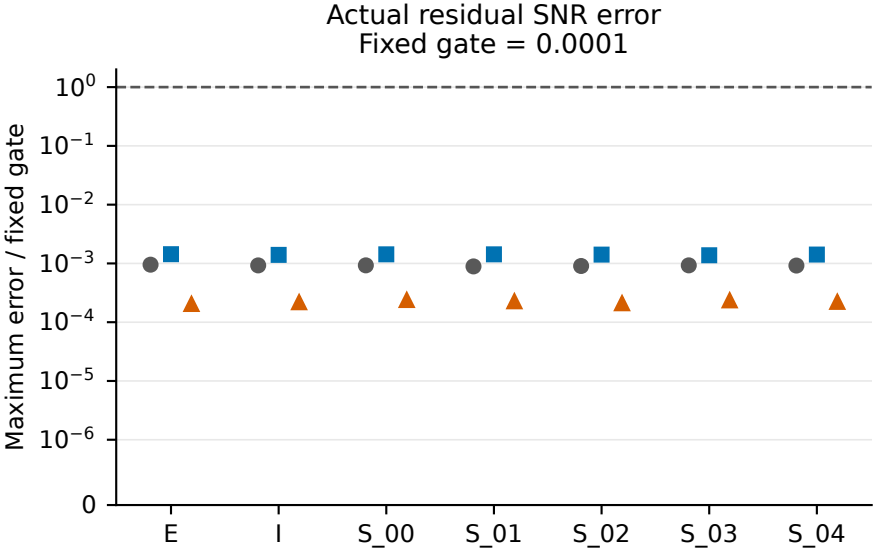


Input validation | measured errors and exact designed invariants

● 10 dB ■ 5 dB ▲ 0 dB



Each point is the maximum over 2,158 ECG records and 6 fixed noises for that condition/SNR (1,877 unique patients); not an uncertainty estimate.
Top: post-addition float64 residual measurements; dashed line = fixed gate. RMS is worst lead; PSD is worst bin difference / that lead's reference peak, then worst lead.
Bottom: designed float32 sign and per-lead PSD identities are exactly zero. Independent RMS has no signed/PSD identity requirement (N/A).
All seven conditions and 10/5/0 dB are retained; fixed horizontal offsets separate SNRs. Zero stays zero on the upper symlog axes.